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PROJECT REPORT: AUTONOMOUS HYBRID SOLAR STREET LIGHTING WITH LOCALIZED FAULT DIAGNOSTICS

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TEAM NAME: TECH TITANS

TARGET PLATFORM: ESP32 MICROCONTROLLER UNIT (MCU)

DOCUMENT TYPE: FULL PROJECT DOCUMENTATION & SOURCE CODE

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1. PROJECT OVERVIEW & VALUE PROPOSITION

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The Autonomous Hybrid Street Light is an intelligent, decentralized, and highly cost-effective smart city infrastructure prototype built around the ESP32 MCU. Modern automated street lights either run continuously on flat-rate grid power or require highly expensive, cloud-connected central management servers to process physical diagnostic data.

This project bridges the gap by implementing "Edge Computing." By shifting the decision-making logic directly to the localized microcontroller, this pole can manage its own power routing, optimize its dimming efficiency via infrared motion tracking, and cross-examine environmental sensors to diagnose physical hardware faults (such as structural damage, loose wiring, or severe lens dust) entirely on its own without requiring a continuous server data plan or expensive dedicated current-sensing chips.

2. SYSTEM COMPONENTS & ELECTRICAL CONNECTIONS

The hardware setup utilizes an array of low-cost sensors arranged to interface safely with the 12-bit Analog-to-Digital Converter (ADC) input rails of the ESP32.

Pin Layout Blueprint:

- GPIO 34 (Analog Input): Ambient Light Sensor (LDR via a pull-down resistor)
- GPIO 35 (Analog Input): Lithium-Ion Battery Monitor (via a 10k/10k divider network)
- GPIO 32 (Analog Input): Solar Panel Voltage Sensing (via a 3x 10k divider network)
- GPIO 2 (Digital Input): PIR Motion Sensor
- GPIO 18 (Digital Output): Active-Low Relay Module (Grid/Battery Source Router)
- GPIO 19 & 21 (PWM Outputs): Dual High-Current Sink Pins bound to the LED load

The 3-Resistor Safety Voltage Divider Network:

To feed a high-voltage solar panel (7V–8V peak output) into a sensitive ESP32 pin (which has an absolute safety ceiling of 3.3V), a unique series network of three identical 10k-ohm resistors is deployed. Tapping the middle junction of the 2nd and 3rd resistors ensures the MCU only reads exactly 1/3 of the true voltage, safely bounding the peak 8V input down to a safe 2.66V maximum window.

```
[Solar Panel (+) ] ───────► [10kΩ Resistor 1]
                       │
                       ├─────► Node A (DO NOT USE: Carries 2/3 Voltage
                                 ~5.3V)
                       │
                 [10kΩ Resistor 2]
```

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                     |
                     ├────► Node B (CONNECT TO GPIO 32: Carries 1/3
                              Voltage ~2.66V)
                     |
              [10kΩ Resistor 3]
                     |
[Common GND ] ◄┴────────── Connect Solar Panel (-) here!
```

3. INDUSTRIAL COST-EFFECTIVENESS & ENERGY CALCULATION

Assuming a standard 30W LED lamp operating for 12 hours every night (6:00 PM to 6:00 AM) at a standard Indian municipal utility rate of ₹8 per unit (kWh):

Traditional Grid-Tied Street Light:

- Continuous draw: 30W x 12 Hours = 360 Wh = 0.36 kWh per day.
- Annual consumption: 0.36 kWh x 365 Days = 131.4 kWh per pole.
- Annual utility cost: 131.4 Units x ₹8 = ₹1,051.20 per pole/year.

The Tech Titans Smart Hybrid Configuration:

- Motion Dimming Impact: Standard operational windows assume 3 hours of active traffic

  (100% Brightness = 30W) and 9 hours of empty streets (30% Dim State = 9W).

  (30W x 3 Hours) + (9W x 9 Hours) = 90Wh + 81Wh = 171 Wh = 0.171 kWh per day.

- Solar Integration Impact: In a typical 365-day year, India experiences roughly 280 sunny/clear days where the battery bank completely handles the 171 Wh nightly load.

  During the remaining 85 rainy/monsoon days, the grid fallback handles 2/3 of the cycle.

- Annual grid consumption: 85 days x 0.114 kWh = 9.69 kWh per pole.
- Annual utility cost: 9.69 Units x ₹8 = ₹77.52 per pole/year.

METRIC	TRADITIONAL GRID	HYBRID SAVINGS
Power Consumption (Pole/Year)	131.4 kWh	9.69 kWh [Less Grid Strain]
Electricity Cost (Pole/Year)	₹1,051.20	₹77.52 [₹973.68 Saved/Pole]
Total Annual Bill (100 Poles)	₹1,05,120	₹7,752 [₹97,368 Saved]

4. SYSTEM LOGIC & FIRMWARE WATCHDOG STRATEGY

The software architecture is governed by a state machine that reads data across a non-blocking timeline using ‘millis()’ instead of processing-stalling ‘delay()’ windows.

Daylight Control (LDR Hysteresis):

The system prevents light flickering during erratic cloud movements or twilight by enforcing a wide hysteresis buffer zone. The light turns ON only when the ambient reading drops below 800 (DUSK), and stays locked OFF once it climbs firmly past 1500 (DAWN). Between 800 and 1500, it safely memorizes its current state.

The Cross-Referenced Weather Truth Matrix:

Standard diagnostics throw false alarms when clouds block solar generation. To solve this, the Tech Titans firmware runs an environmental cross-examination:

1. If the panel voltage is high (>= 6.5V) during the day, the system logs normal operation.
2. If the LDR detects a blindingly bright sky (> 3000) but the solar voltage drops below 4.5V, the "Trap is sprung". The sky is bright, yet the panel is producing nothing.

   The ESP32 safely trips a CRITICAL HARDWARE ALARM, diagnosing a physical failure.

3. If it is daylight but the LDR is moderately low, the system assumes cloud/storm cover and automatically suppresses the alarm, preventing false diagnostic tickets.